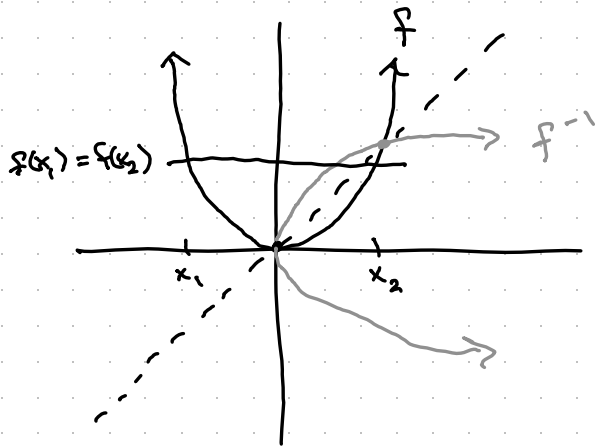


One-to-one functions - 0.1.4

When can inverses fail to exist?

Example: Consider $f(x) = x^2$.



Problem: We know what f^{-1} should look like if it exists, but this fails the vertical line test (it isn't really a function).

→ A function has one output per input, and we check this with the vertical line test.

We can check whether an inverse will exist using the horizontal line test.

Definition: Call a function f one-to-one if for all x_1 and x_2 satisfying $f(x_1) = f(x_2)$, we have $x_1 = x_2$. Equivalently, f is one-to-one if whenever $x_1 \neq x_2$, then also $f(x_1) \neq f(x_2)$.

(Both say that f passes the horizontal line test.)

Theorem: If f is one-to-one, then f is invertible (that is, f^{-1} exists).

If f fails to be one-to-one, often we can pick a smaller domain to work with.

Example: $f(x) = x^2$ is not 1-1 on $(-\infty, \infty)$
(e.g. $f(-1) = f(1)$, but $-1 \neq 1$)

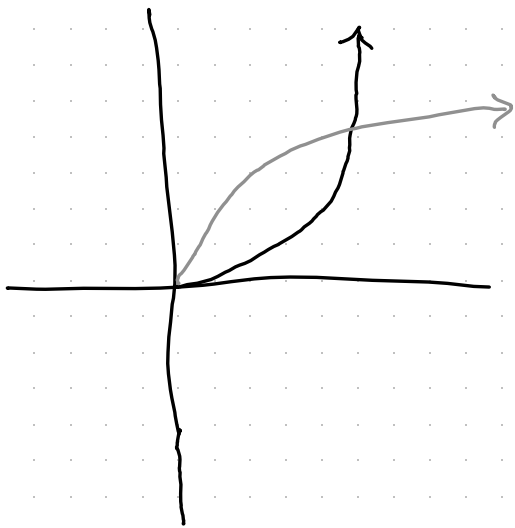
However, $g(x) = x^2$ on the domain $[0, \infty)$ is 1-1.

So this function g
has an inverse!

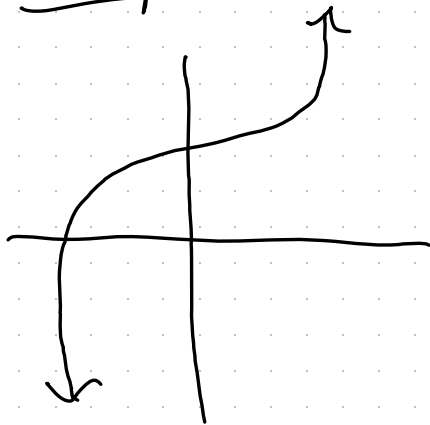
$$g^{-1}(x) = \sqrt{x}$$

$h(x) = x^2$ on domain $(-\infty, 0]$
is also 1-1 with inverse

$$h^{-1}(x) = -\sqrt{x}.$$



Example: Consider $f(x) = x^3 + 4$. Is f 1-1?



passes the HLT ✓

Algebraically, we could suppose x_1 and x_2 are inputs satisfying $f(x_1) = f(x_2)$, and try to show that $x_1 = x_2$.

$$x_1^3 + 4 = x_2^3 + 4$$

$$x_1^3 = x_2^3 \quad (\text{subtracting 4})$$

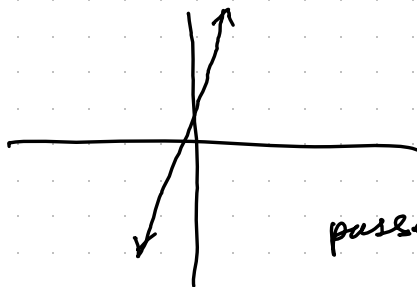
$$x_1 = x_2 \quad (\text{taking roots})$$

Example: Show that $f(x) = 7x + 1$ is 1-1.

$$7x_1 + 1 = 7x_2 + 1$$

$$7x_1 = 7x_2$$

$$x_1 = x_2$$



passes the HLT.